MISP Galaxy in 2.5

Explore the knowledge base, assess context, share carefully

CIRCL / MISP Project

https://www.misp-project.org/

Reference snapshot: 30 October 2026



Find context, assess evidence, make an assertion

- Explore the knowledge base beyond actor names.
- Choose the right cluster and attachment scope.
- Interpret relations and matrices carefully.
- Practise discovery, custom knowledge and contribution.

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Reference MISP 2.5.50 and misp-galaxy 285d1477, reviewed 30 October 2026. Threat-actor UI captures were made on 1 November 2026 in a disposable 2.5.50 lab using that same Galaxy revision. Allow 40–50 minutes plus a workshop. Counts and UI capabilities depend on revision, permissions and settings. The README pins the sources and supplies reproducible commands.

Explore the knowledge base

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Measured from the pinned checkout by the supplied offline explorer. These are entries and edges, not counts of distinct threat actors, verified claims or records visible to every MISP user. Collections can overlap and reuse UUIDs, including different ATT&CK views. Pin the revision when comparing results.

Start with the public catalogue

- Browse `https://www.misp-galaxy.org/`.
- Follow JSON and original-source references.
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The public knowledge-base website does not require a MISP instance. The source repository is the reproducible machine-readable reference. A local MISP installation can have a different database/import revision from the website or your checkout; distinguish those when debugging missing knowledge.

Choose a family that answers the question

- ATT&CK: observed techniques, software and activity groups.
- D3FEND and preventive measures: defensive context.
- DISARM: information-manipulation techniques and responses.
- Country, sector, detection and other domain knowledge.

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The repository covers substantially more than actor and malware lists. Prefer domain-specific ATT&CK collections when appropriate; the combined collection contains broad and historical material. The README links selected catalogue families rather than embedding an unmaintainable exhaustive list.

Galaxy, cluster and element

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The repository has galaxies/name.json definitions and clusters/-name.json collections. A collection contains a values array of individual cluster entries. Its UUID is not the UUID of each entry. Family and collection are associated by type; do not assume their UUID fields are interchangeable.

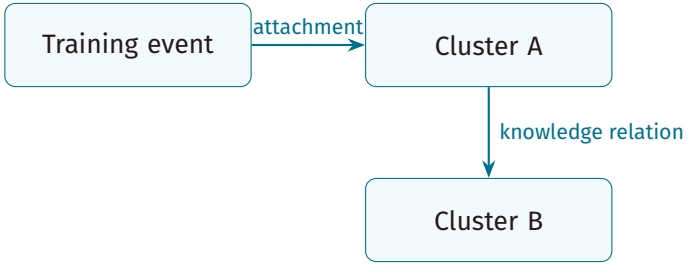
Context is different from incident data

- Taxonomy: classification vocabulary.
- MISP object: grouped typed attributes.
- Galaxy cluster: structured knowledge attached as context.

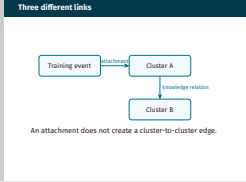
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Attaching an actor cluster is an analytical assertion, not a technical indicator or proof of causation. Object references, value correlations and analyst-data relationships have different purposes. Use the taxonomy and object-template courses for those models.

Three different links



An attachment does not create a cluster-to-cluster edge.



A cluster-to-cluster edge describes knowledge independently of a particular event. An event association says the selected concept is relevant to that event. Taxonomy tags can qualify a relationship without being the relationship's identity. Source, direction and evidential scope need separate review.

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Synonyms support discovery; they do not prove a one-to-one mapping between providers. Attribution labels can overlap or be disputed. The explorer returns candidates, not an entity-resolution decision. Use original sources, scope and uncertainty to choose among candidates.

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Historical knowledge can be valuable without representing the current preferred concept. A successful search is not proof that an entry is active. The combined ATT&CK collection includes historical material, so the workshop uses an enterprise-specific collection and an external identifier.

Apply context in MISP 2.5

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- Attribute scope can express a narrower observation.
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Parent context can appear in attribute exports without being directly attached at attribute level. Current bulk attachment supports optional attribute-to-event mirroring when enabled; that broader scope can change the assertion. Explain why the whole event supports the concept before mirroring.

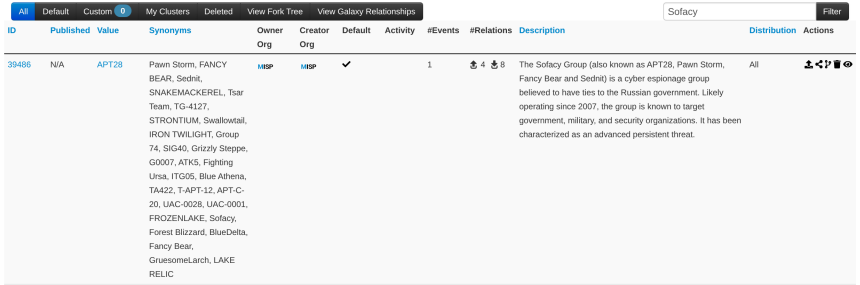
Use the installed cluster metadata

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Imported default tag names commonly look like misp-galaxy:type="value". Custom knowledge and naming conventions need not match every imported default. Use the actual metadata supplied by the instance rather than treating a guessed tag string as a complete, accessible cluster definition.

Threat Actor: search for Sofacy



The APT28 result lists Sofacy as a synonym. Inspect the candidate.

MISP 2.5.50; synthetic lab, 1 November 2026.



Open Galaxies, select Threat Actor and filter for Sofacy. This cropped live capture shows the first result. The actual UI search also returned Snake Wine and White Bear because their descriptions mention Sofacy; unlike the offline explorer, this UI search includes descriptions. A search hit is not an identity-equivalence or attribution decision. Click APT28 to inspect it. Do not use installation-specific numeric IDs as portable identities. Capture provenance and full navigation steps are in screenshots/README.md.

Inspect the cluster identity and provenance

Threat Actor :: APT28

Cluster ID	39486
Name	APT28
Parent Galaxy	Threat Actor
Description	The Sofacy Group (also known as APT28, Pawn Storm, Fancy Bear and Sednit) is a cyber espionage group believed to have ties to the Russian government. Likely operating since 2007, the group is known to target government, military, and security organizations. It has been characterized as an advanced persistent threat.
Default	Yes
Version	357
UUID	5b4ee3ea-eee3-4c8e-8323-85ae32658754
Collection UUID	7cdff317-a673-4474-84ec-4f1754947823
Source	MISP Project
Authors	Alexandre Dulaunoy, Florian Roth, Thomas Schreck, Timo Steffens, Various
Distribution	All communities
Owner Organisation	MISP
Creator Organisation	MISP
Connector tag	misp-galaxy:threat-actor="APT28"
Events	1 event
Attributes	1 attribute

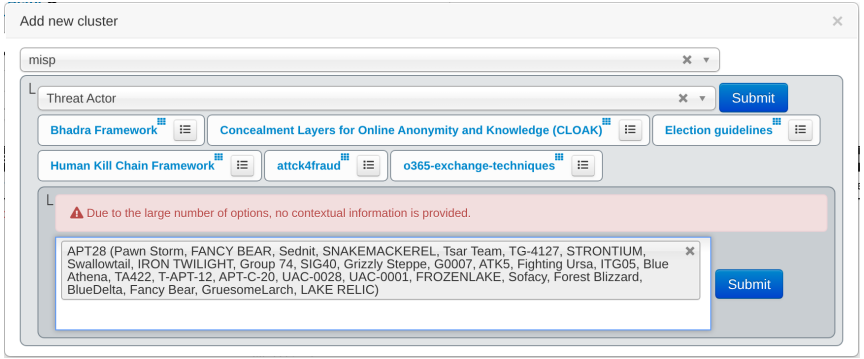
Check the cluster UUID, collection, version and source.

MISP 2.5.50; synthetic lab, 1 November 2026.



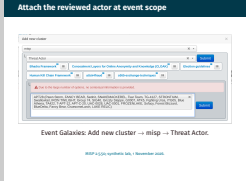
This is the Threat Actor APT28 cluster, UUID 5b4ee3ea-eee3-4c8e-8323-85ae32658754, collection version 357. Distinguish its UUID from the collection UUID and from the instance-specific numeric cluster ID. The connector tag is misp-galaxy:threat-actor="APT28". Below this cropped identity panel, inspect the paginated metadata or JSON view for synonyms and original refs, including the MITRE G0007 reference. Review what those sources actually support. The country-of-origin attribution-confidence metadata is not confidence in this synthetic event attachment.

Attach the reviewed actor at event scope



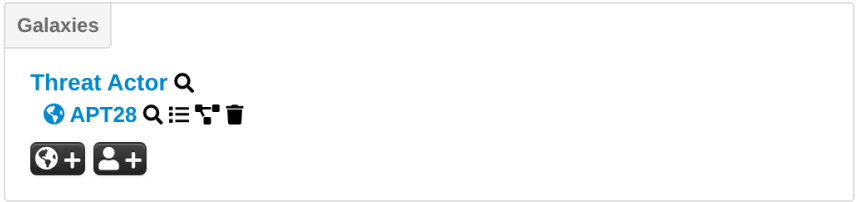
Event Galaxies: Add new cluster → misp → Threat Actor.

MISP 2.5.50; synthetic lab, 1 November 2026.



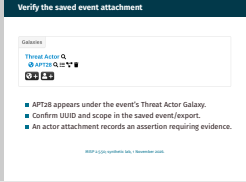
Use an unpublished organisation-only synthetic event. In the event Galaxies panel, choose Add new cluster, the misp namespace, Threat Actor, then search for and select APT28. The final Submit attaches the selected cluster; earlier namespace/family controls only navigate the picker. This large collection displays a warning that contextual information is omitted, so inspect the separate cluster detail view before selection. Add new local cluster is a different association-scope choice. Attribute-level controls are separate. This capture precedes the final Submit.

Verify the saved event attachment



- APT28 appears under the event's Threat Actor Galaxy.
- Confirm UUID and scope in the saved event/export.
- An actor attachment records an assertion requiring evidence.

MISP 2.5.50; synthetic lab, 1 November 2026.



Reload the event after saving. This focused capture shows the event Galaxies panel, not an attribute-level control. The live API check confirmed the expected UUID and connector tag at event scope, a non-local association, and no direct actor attachment on the comment attribute. The event remained unpublished and organisation-only: a non-local association does not make its event public. Use the magnifying-glass action for cluster details. This lab illustrates a cited-report assertion; it is not a real incident or independent attribution. Detach the training association after the exercise. No peer sync, fork publication or cross-role visibility test is claimed.

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- Custom-cluster distribution controls knowledge access.
- Local attachment controls the association scope.
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A local-only Galaxy also restricts non-local attachments. Attaching a private cluster does not automatically grant recipients access to its definition. The API checks both target permissions and cluster visibility. A cluster relation and an event attachment are different operations with different permissions.

Matrix highlights are observations, not coverage

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- Cluster kill-chain metadata places entries in that layout.
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The galaxy kill_chain_order and cluster meta.kill_chain must use matching keys/prefixes. Interpret the selected domain/platform and source revision. Separate observed technique, available detection, tested detection and effective prevention. A matrix is not an attribution-confidence or maturity score.

Read graph edges as sourced assertions

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- Review confidence or handling tags where supplied.
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Relations may span collections and point to concepts absent from a restricted local view. An unresolved target does not establish identity equivalence. Do not copy every knowledge-base relation into the evidential claims of an incident.

Custom clusters and forks are supported

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- Fork a default cluster for local analysis.
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Default imported clusters are protected from normal edits. The reviewed fork workflow records extends_uuid and extends_version while creating separate custom identity. Forking is not automatic agreement with future upstream changes. Normal editing and publication require appropriate permissions, including Galaxy-editor privileges where applicable.

Updating files is different from updating the catalogue

- Select and update the intended Git revision.
- Import/update the MISP database through the operator workflow.
- Review changed knowledge and custom forks before sharing.

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The model reads app/files/misp-galaxy. Admin updateGalaxies imports from disk; it does not fetch Git. Avoid force/wipe operations as a routine exercise. Custom cluster synchronisation depends on publication, permissions, sharing groups and peer settings; forwarding a tag alone is not complete knowledge distribution.

Explore, validate and contribute

Explore a pinned checkout offline

```
python3 examples/explore.py /tmp/misp-galaxy-lab --stats
python3 examples/explore.py /tmp/misp-galaxy-lab \
  --search T1059 \
  --collection mitre-enterprise-attack-attack-pattern \
  --limit 3
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Clone and check out the reviewed revision first; full commands are in the README. The explorer is standard-library Python, needs no API key and makes no network calls. It searches value, synonyms and external_id, with a result bound. It prints provenance and raw metadata to support source review.

Repository structure: family versus collection

```
galaxies/training-knowledge.json
  type, namespace, name, uuid, version, description

clusters/training-knowledge.json
  type, name, uuid, version, source, authors, category
  values: [individual cluster entries]
```

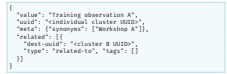

Repository structure: family versus collection
galaxies/training-knowledge.json
type, namespace, name, uuid, version, description
clusters/training-knowledge.json
type, name, uuid, version, source, authors, category
values: [individual cluster entries]

This is a file-layout illustration, not an API payload. The supplied galaxy.json and clusters.json are valid standalone repository-format examples. A private MISP cluster API export can have a different envelope. Match family and collection type and preserve the distinction between family, collection and individual identities.

An individual cluster can reference another

```
{
  "value": "Training observation A",
  "uuid": "<individual cluster UUID>",
  "meta": {"synonyms": ["Workshop A"]},
  "related": [{
    "dest-uuid": "<cluster B UUID>",
    "type": "related-to", "tags": []
  }]
}
```


An individual cluster can reference another



The angle-bracket UUIDs illustrate the structure and must not be submitted as data. The executable fixture contains real generated UUIDs. Relationships need meaningful types, correct direction and resolvable targets under an explicit policy. Schema success alone does not establish all those semantics.

Validate more than JSON syntax

- Check both repository schemas.
- Check identities, matching types and relationship targets.
- Exercise negative cases and inspect original sources.

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The verifier checks the synthetic fixture and four negative cases: mismatched type, dangling target, duplicate fixture UUID and blank label. It also performs a real ATT&CK lookup and tests empty/unknown collection cases. Fixture references are internal; real cross-collection knowledge needs a broader target-resolution policy.

Workshop: find, assess, attach

- Find a technique by external identifier and read its evidence.
- Compare a synonym search and a domain-specific collection.
- Attach to a disposable event; compare attribute and local scope.
- Inspect matrix/export results and remove training data.

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The optional live lab uses an unpublished organisation-only event and an authorised user. Follow the captured Threat Actor workflow for APT28, then compare a technique attachment. Fork only when permitted, keep custom knowledge restricted and compare visibility with a second ordinary user. The captured 2.5.50 lab verified only actor discovery, cluster inspection and event attachment with an API scope check; the other live exercises are optional and were not executed for this update.

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- Manual collection: edit JSON with evidence and provenance.
- Generated collection: inspect the generator and upstream source.
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CONTRIBUTE.md distinguishes generated ATT&CK knowledge from manually maintained collections. Prefer original references. For threat-actor data, cfr-* metadata needs CFR evidence and attribution-confidence has a specific country-of-origin meaning. Do not treat it as universal confidence in an incident attachment.

Follow the current upstream validation workflow

```
./jq_all_the_things.sh
# Review and commit intended normalised changes.
./validate_all.sh
```


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Review and commit intended normalised changes.
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Use a dedicated checkout with jq, moreutils/sponge and the current Python validation dependencies. Formatting can rewrite many files. The reviewed CI also uses PyMISPGalaxies tests and compares the generated README index. The course verifier is not a substitute for that complete pipeline.

Make knowledge reusable and reviewable

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The earlier lesson, images and published PDF remain archived outside standard training. The new course points to the large live knowledge base and pins the reviewed snapshot for reproducibility. Repository counts and schema checks do not prove the truth or completeness of every entry.